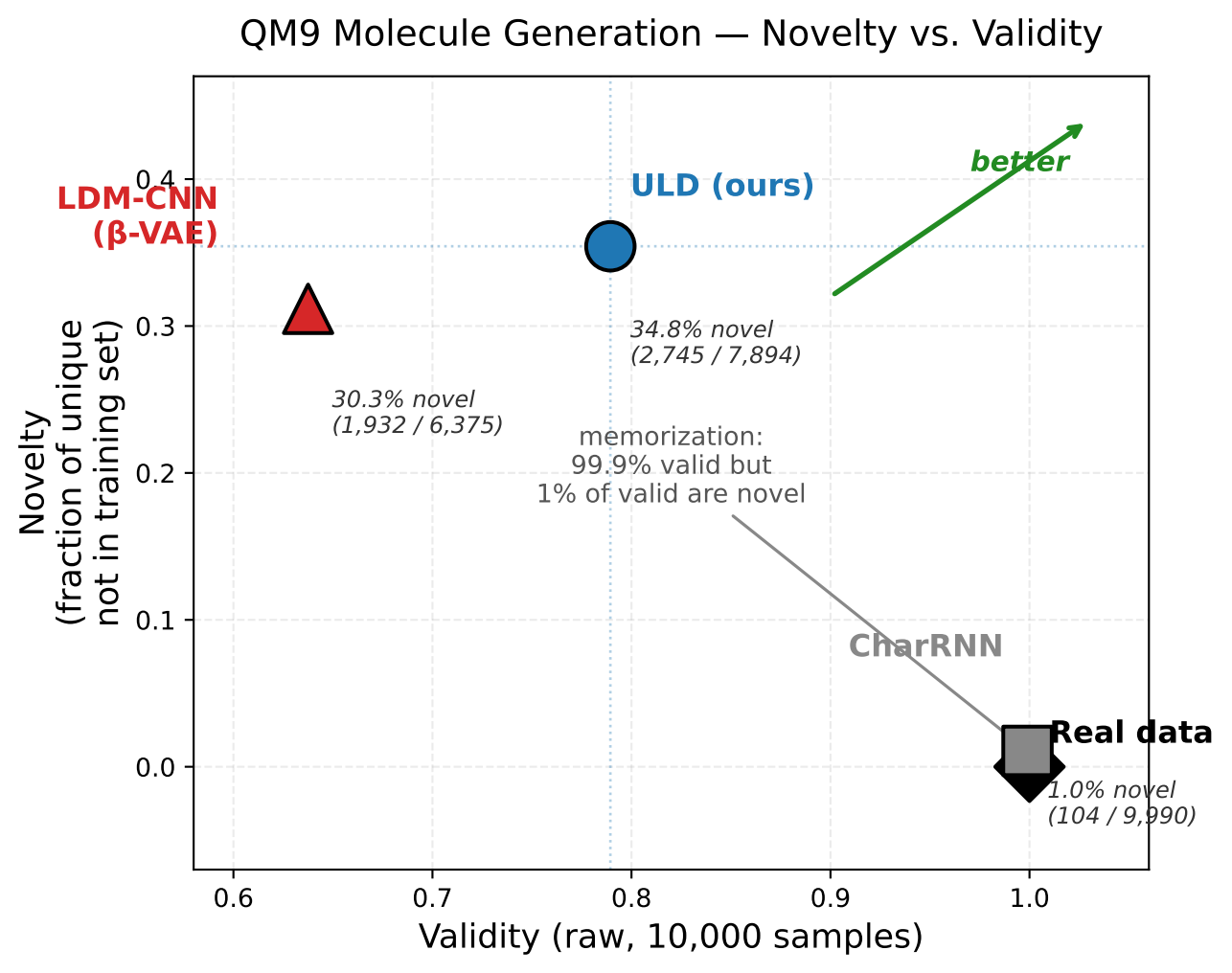
QM9 Molecule Generation — Novelty vs. Validity
LDM-CNN
(β-VAE)
ULD (ours)
better
34.8% novel
(2,745 / 7,894)
30.3% novel
(1,932 / 6,375)
memorization:
99.9% valid but
1% of valid are novel
CharRNN
Real data
1.0% novel
(104 / 9,990)
Novelty
(fraction of unique
not in training set)
0.0
0.1
0.2
0.3
0.4
0.6
0.7
0.8
0.9
1.0
Validity (raw, 10,000 samples)